SDG BLOCKCHAIN ACCELERATOR

Market Penetration and Scaling Strategy

Plastiks x UNDP India

This document outlines the market penetration strategy, branding approach, user acquisition plan, go-to-market strategy, and long-term scaling roadmap for the Plastiks x UNDP India pilot developed within the SDG Blockchain Accelerator program.

1. Project Overview

The Plastiks x UNDP India pilot addresses India's plastic waste challenge by bridging the gap between informal waste collection and formal recycling data. In India, an estimated 80% of plastic collection relies on the informal sector, with over 10,000 waste pickers per city on average. These waste pickers provide essential environmental services but remain largely invisible in official waste management systems. The pilot aims to "mainstream" informal waste pickers into the plastic recycling value chain, using blockchain technology and digital tools to record their contributions and reward their efforts. By giving waste workers a QR-coded digital identity (even without smartphones) and logging each kilogram of recovered plastic on an immutable ledger, Plastiks creates a trusted, transparent traceability system. This system generates verifiable data on plastic recovery, preventing fraud and double-counting while ensuring that every kilo of plastic collected is accounted for on-chain. The ultimate goal is to transform plastic waste from an untracked liability into a source of verified environmental impact that supports policy, compliance, and community livelihoods.

How the Platform Works

When a waste picker delivers recyclable plastic to a collection center or recycler, their unique QR code is scanned to log the transaction. No smartphone is required for the waste picker. They can carry a simple printed QR ID card or tag. A field operator or shop owner uses a mobile app or point-of-sale device to scan the code and record the material type and weight. Each sack or batch of collected plastic can be tagged with a QR code linked to the picker and transaction, ensuring traceability through the recycling process. The data (including timestamp, location, picker ID, and photos of the material if available) is then uploaded to the Cardano blockchain as a tamper-proof record. This process creates a Digital Recovery Certificate for each verified recovery action. The accumulation of these certificates can be monetized into Plastic Credits, representing volumes of plastic verified as collected and processed. These credits serve as evidence of recycling for companies with Extended Producer Responsibility (EPR) obligations and as impact metrics for CSR/ESG reporting. A real-time dashboard aggregates the data, offering

government authorities, brands, and NGOs an auditable view of plastics recovery aligned with SDG indicators (notably SDG 12.5.1 on national recycling rates, SDG 8.3.1 on formalizing informal work, and SDG 13.2.1 on integrating climate measures) by quantifying waste reduction, livelihood enhancement, and carbon-related benefits.

Who Benefits

The pilot's inclusive design means all stakeholders in the ecosystem benefit. Waste pickers gain formal recognition through digital IDs and potentially better compensation (e.g. through credit revenues or performance-based incentives) for the estimated millions of tons of waste they divert from landfills. Local recyclers and aggregators gain access to a reliable tracking system and a broader supplier base, improving their input quality and enabling compliance with ethical sourcing standards. Brands and Producers (the EPR stakeholders) benefit from credible, blockchain-verified data to meet government mandates and avoid greenwashing – they can confidently report how much plastic they have helped remove from the environment, backed by immutable certificates. Government bodies and urban local authorities (ULBs) benefit from improved waste management outcomes and data for policy-making: the system helps enforce EPR rules, informs city waste planning, and integrates with national digital infrastructure (such as the centralized EPR portal of the pollution control board) for tracking recycling rates. Finally, the environment and society at large benefit from reduced plastic leakage, cleaner communities, and improved livelihoods for waste workers. By design, the pilot aligns with UNDP priorities on inclusive growth and environmental sustainability, demonstrating how a blockchain-based solution can advance the circular economy in tandem with social protection initiatives (e.g. linking waste picker IDs to government schemes like health insurance or the NAMASTE helpline for sanitation workers).

The following sections detail our strategies and plans across the key areas of market penetration, branding, user acquisition, go-to-market execution, and post-accelerator sustainability and scaling. Each section explains how the pilot's approach is tailored to the Indian context and the SDG Accelerator's objectives, with a cohesive narrative of how the project will grow from pilot to impactful platform.

2. Market Penetration Strategy

India's waste management landscape presents a vast but fragmented market. Our market penetration strategy focuses on two interconnected fronts: (1) Bottom-up integration of the informal waste picker network into the Plastiks platform, and (2) Top-down engagement with institutional partners (municipal bodies, recycling industries, and brand owners) who drive demand for verified recycling data. By uniting these fronts, Plastiks aims to penetrate the market as the go-to solution that links grassroots waste collection with high-level compliance needs.

2.1 Informal Sector Integration

Given that informal workers handle the majority of recyclable plastics in India, onboarding waste pickers and small scrap shops is critical for coverage. The pilot strategically targets major recycling hubs identified by UNDP, for instance, urban areas in Pune (Maharashtra) and the NCR (Noida/Greater Noida), where large concentrations of waste pickers and aggregators operate. In these locales, the project works through local NGOs and cooperatives that have trust within the waste picker community. By partnering with organizations like the Alliance of Indian Wastepickers and local cooperatives (e.g., SWaCH in Pune), we gain grassroots credibility and access to networks of workers who can be enrolled en masse. Each waste picker or scrap dealer joining the platform is provided a QR code ID card and training on the simple process (e.g. ensuring they present their code whenever they sell materials to a registered buyer). This low-tech approach (print cards and basic phone scanning) removes barriers like smartphone ownership or digital literacy, making participation feasible for even the most marginalized workers. The promise to waste pickers is clear: formal recognition and potential financial upside. As they log collections, they build a verifiable work record that could support stable pricing for their materials and even unlock micro-rewards (for example, a bonus per kilogram via a sponsor's CSR fund, or eligibility for government welfare schemes once registered). By structurally improving their livelihoods and status, we anticipate strong word-of-mouth recruitment – waste pickers have incentive to encourage their peers to join, accelerating adoption organically.

2.2 Institutional and Corporate Engagement

In parallel, Plastiks penetrates the market from the top by aligning with the pressing needs of regulators and producers. India's new Plastic Waste Management rules (2022 and upcoming 2025 mandates) require producers, importers, and brand owners to not only recycle a portion of their plastic but also to digitally track and report the waste recovery via a centralized system. Plastiks positions itself as a ready-made solution to help companies and municipalities meet these requirements. Our strategy involves close collaboration with ULBs and state pollution control boards in pilot regions to incorporate Plastiks data into official reporting pipelines. By demonstrating how our tamper-proof, blockchain-based tracking can feed into the national EPR compliance portal (with each kilogram recovered verifiably linked to a source and outcome), we make the platform indispensable for meeting government benchmarks. We also engage directly with major brand owners and industry associations (such as FICCI's circular economy initiative) to advocate Plastiks as part of their EPR action plans. The unique value proposition – verifiable proof of recycling that withstands audit – attracts companies keen to avoid penalties and reputational risk under tighter regulations. By initially offering pilot partnerships (e.g. allowing a large consumer goods company to sponsor the collection of, say, 50 tons of plastic through Plastiks in one city, and then use the data for their compliance report), we seed demand among corporates. As these early adopters publicize their success ("X tons of plastic recovered with full traceability in partnership with Plastiks and UNDP"), it builds market momentum and pressure on competitors to also participate or purchase Plastic Credits to offset their footprints.

2.3 Leveraging Policy and Ecosystem

A supportive policy environment amplifies our market penetration. The pilot aligns with governmental and UNDP initiatives to formalize waste labor and increase recycling. For example, India's Ministry of Housing and Urban Affairs has promoted integrating informal workers into municipal systems since the Solid Waste Management Rules 2016. By working within such frameworks, we ensure Plastiks is not seen as an outside intervention but as a tool that complements public programs like the NAMASTE scheme (which provides IDs and welfare linkages for waste workers). Indeed, Plastiks can be presented as helping achieve government targets (such as

enrolling waste pickers into social protection and documenting their economic contributions). Moreover, our partnership with UNDP India gives credibility and access at high levels – UNDP's convening power allows us to engage multiple city governments, ministries, and corporate leaders through roundtables and knowledge events. These forums not only raise awareness but position Plastiks as an integral part of the national circular economy strategy, thereby smoothing entry into new markets (cities or states) as a trusted, mission-aligned platform.

In summary, the market penetration strategy for the pilot is to deeply embed into the existing ecosystem: empowering the base of waste collectors to join and benefit, while reassuring institutions and companies that Plastiks fulfills their needs for compliance, ESG, and SDG impact. Success in the initial target hubs will generate case studies and champions (among both waste picker groups and corporate/government partners) that we will leverage to replicate the model across other urban centers. By aligning economic incentives (financial rewards, compliance cost savings) with social and environmental outcomes, Plastiks will penetrate the market as a sustainable, network-effect-driven solution where each new participant increases value for all.

3. Branding and Positioning

Our branding strategy for the Plastiks x UNDP India pilot emphasizes trust, inclusivity, and innovation, crafting a narrative and identity that resonate with both grassroots users and high-level stakeholders. Plastiks is positioned not merely as a tech product, but as a social impact platform at the intersection of sustainability and livelihood empowerment. Given the diverse audience (waste pickers, NGOs, government officials, corporate CSR and ESG teams), our brand messaging remains consistent in core values while adapting in tone for each segment.

3.1 Core Brand Identity

At its heart, Plastiks stands for transparency, accountability, and empowerment. The use of blockchain is a key differentiator that we highlight in branding as a symbol of trust and permanence, for instance, messaging that "every kilogram recovered is forever recorded and cannot be tampered with" underlines a commitment to truth in

environmental reporting. We avoid techno-jargon in public-facing materials; instead of promoting "blockchain" itself, we emphasize what it enables: *"Proof, not promises"* in sustainability claims, and *"Invisible heroes made visible"* in reference to waste pickers. The visual branding in the pilot uses simple, relatable imagery, e.g. a waste picker with a QR code badge and a sack of plastic turning into a data point on a screen, to convey how traditional work links to modern data. We incorporate the logos of UNDP and the SDG wheel in co-branding to reinforce institutional credibility and the sense that this pilot is part of a larger global mission (the SDG Blockchain Accelerator). This not only lends authority when approaching officials or companies, but also instills pride and hope in waste picker communities that this initiative is backed by the United Nations and aimed at achieving the Sustainable Development Goals.

3.2 Localized and Inclusive Approach

Branding is carefully localized for the Indian context. While the platform name "Plastiks" is used consistently, we ensure materials (posters, flyers, training guides) are bilingual (English and local languages such as Hindi or Marathi in pilot areas) and culturally sensitive. For example, in community meetings we might refer to the platform as a "digital Safai Mitra (cleanliness friend)", linking to government nomenclature for sanitation workers to create familiarity. The tagline for waste picker outreach could be "Paryavaran ke Nayak" (Heroes of the Environment) to celebrate their role and position Plastiks as the vehicle that will showcase their heroism to the world. We also consciously brand the QR code ID cards not as mere tokens but as "Digital IDs for Dignity", possibly including each worker's photo and a line like "Recognized Environmental Service Provider" under the Plastiks and UNDP logos. This gives a sense of pride and belonging, which is a powerful intangible incentive. The branding strategy avoids any appearance of exploitation; rather than focusing on the waste problem alone, it focuses on the solution and the people making it happen. For corporations and governments, we frame Plastiks as an "anti-greenwashing, pro-people" solution – a refreshing brand that stands apart from generic consulting or audit firms. This human-centric aura, combined with cutting-edge tech, makes our brand attractive for companies to associate with (enhancing their own brand by partnership) and for local officials to support publicly.

3.3 Communication Channels and Storytelling

We support the brand through strategic communications. During the pilot, success stories are documented: for instance, the journey of a woman waste picker who, through her QR code logs, shows up as contributing 500 kg of recycling – we turn that into a human story shared on social media, blogs, and local press (with consent and appropriate anonymization if needed). Such storytelling puts a face to the data and reinforces the Plastiks brand ethos that **"impact is about people and planet, measured in kilograms and smiles"**. We will also highlight metrics in a visually engaging way: live counters of plastic recovered, number of workers onboarded, etc., displayed on our dashboard and in communications to emphasize progress toward SDGs (e.g., "X tons recovered, contributing to SDG 12.5 and 13.2"). The brand voice remains professional but optimistic and solution-oriented, mirroring UNDP's style of highlighting positive change. Additionally, we maintain consistency in design – using the SDG color palette and Plastiks' logo on all materials to build recognition.

3.4 Collaborative Branding with Partners

As a pilot under the SDG Accelerator, co-branding is key. In all public presentations, we use "Plastiks x UNDP India" to acknowledge the partnership. We also incorporate the names/logos of local partners (municipalities, NGOs, or corporate sponsors) where appropriate to foster a sense of collective ownership. This multi-logo approach in branding events or reports shows that Plastiks operates within an ecosystem – reinforcing trust, as stakeholders see endorsements from entities they already respect. For instance, an inaugural pilot event might have banners featuring the city government seal alongside Plastiks and UNDP, signaling public-sector buy-in. For companies, we offer a chance to co-brand specific campaigns (e.g., a brand that funds plastic credits can be acknowledged as "Plastic Neutral Champion – powered by Plastiks"). This strategy turns branding into a value proposition for user acquisition as well, by leveraging partner reputations and giving them visibility.

In sum, our branding for the pilot is polished, empathetic, and impact-driven. We present Plastiks as a catalyst for circular economy transformation that everyone, from a waste picker in Pune to a CEO in Mumbai, can trust and take pride in. The consistency and clarity of this brand will support all other aspects of the project by

building recognition, credibility, and emotional connection, which are crucial for formal milestone review and for real-world adoption alike.

4. User Acquisition Strategy

The user acquisition strategy for Plastiks x UNDP India is designed to rapidly onboard participants on both the supply side (waste pickers, scrap aggregators, recyclers) and the demand side (enterprises and institutions that utilize the data/credits). Recognizing that this is a multi-sided platform, we have tailored approaches for each user segment, focusing on ease of adoption, clear incentives, and trust-building.

4.1 Waste Pickers and Field-Level Aggregators

Acquiring users in the informal segment requires a high-touch, community-based approach. We begin by conducting sensitization workshops in pilot communities, often in collaboration with local NGOs or waste picker cooperatives. These sessions introduce the concept in simple terms: for example, explaining that each person's work can be recorded and acknowledged through a "Chinna (token)", a small QR code badge, and that by participating they may earn a more stable income over time. We emphasize that there is no cost or risk to them – the platform does not charge waste pickers any fees. Instead, it is a tool for them to document their daily collections and get recognized by the municipality and companies as essential service providers. To overcome skepticism (a common issue when introducing new systems to communities that have faced exploitation), we rely on peer champions. In each area, we identify a respected waste picker or scrap dealer and provide them with training to become an ambassador. These ambassadors share their first-hand experience once they start using the system ("I delivered 20 kg and got my receipt on the blockchain; no one can cheat me of that record"). Word-of-mouth via these local influencers has proven effective in similar contexts of community technology adoption. Additionally, to lower barriers, we provide onboarding kits: each registered waste picker gets a durable QR code card, a simple printed pamphlet in their language, and a contact number (helpdesk operated by the project) for any questions. The QR-based digital identity is pivotal – since many waste pickers do not

have smartphones or even formal IDs, being given a scannable ID card confers a sense of identity and belonging. It also means they can start participating immediately without any new device; the scanning is done by others (collection point operators). We coordinate with scrap shop owners and aggregators to ensure they are equipped (with the mobile app and possibly a cheap QR scanner or smartphone provided by the project if needed) and trained to perform scans and data entry. As these small businesses see the benefit – e.g., faster transactions, digital logs to prevent disputes, and possibly better prices from recyclers for "verified" material – they become enthusiastic recruiters as well, urging all their supplier pickers to get on the system (to streamline their own record-keeping).

4.2 Incentives for Retention

Acquiring users is only the first step; we focus equally on retaining them by providing tangible and immediate value. For waste pickers, an important incentive is **financial**. While the long-term vision is that plastic credits create added income, during the pilot we implement a basic reward scheme: for example, for every kilogram logged, the waste picker earns an extra INR 0.50 as a bonus (funded by our CSR partners or grant). This effectively raises their per-kg earning slightly as a thank-you for using the system. Even a small bonus can be significant at scale and helps nurture a habit of participation. We also plan periodic recognition events – e.g., monthly "Leaderboard" announcements where top contributors (in terms of volume or consistency) are acknowledged by local officials or given a small award (safety gear, grocery vouchers, etc.). This social recognition ties into human motivation and encourages others to participate actively. Another element is **feedback and education**: through the QR system, we can send simple feedback messages: if a waste picker logged 100 kg in a month, they might receive a message like "Congratulations, you helped recycle 1000 bottles worth of plastic this month!" – connecting their effort to environmental impact, which can be a source of pride. Over time, as the platform matures, we aim to link these users with other services (financial inclusion, health camps, etc.), but even within the pilot, we coordinate with UNDP's ongoing programs so that registered Plastiks users are on the radar for any **social protection enrollment drives** in the area. This way, joining Plastiks is

associated with being part of a progressive community that gets access to more opportunities.

4.3 Recyclers and Processing Facilities

For recyclers (the next stage in the value chain who buy in bulk from aggregators or directly from waste picker groups), user acquisition focuses on demonstrating operational and market benefits. We target large recycling centers in the pilot regions – often these are already engaged in Extended Producer Responsibility tie-ups and are eager for efficient tracking solutions. We onboard them by integrating our system with their existing weighing and recording processes. For instance, if a recycler receives a truck of plastic, they can use the Plastiks interface to scan all QR-tagged sacks in the load, automatically summing weights per waste picker or source. This provides instant traceability reports which they can use for their own audit and for selling the processed plastic with a verified origin. We assure recyclers that Plastiks will increase their credibility with buyers (some manufacturers prefer or even pay more for recycled plastic that comes with provenance data). Moreover, we address a major pain point: fraud prevention. By using blockchain records and photo evidence, the recycler can avoid scenarios where the same material is claimed twice or non-recyclable material is mixed in, as each entry is verified and time-stamped. To acquire these users, we leverage our institutional partners: UNDP and industry associations introduce Plastiks to recycling industry forums, and we offer free setup and training during the pilot phase. Essentially, we remove any switching cost and emphasize that adopting Plastiks will prepare them for the coming future where traceability is mandated by regulators and buyers (e.g., upcoming requirements for QR/barcode tracking of packages). Early adopter recyclers are likely to become champions that convince others, especially once they experience smoother compliance reporting using our dashboard.

4.4 Brands and Corporate Users

On the demand side, user acquisition means signing up the companies and organizations who will utilize the data or purchase Plastic Credits. For the pilot, rather than open a generic marketplace, we take a curated partnership approach. We have identified a few forward-thinking companies (for example, consumer goods

companies or packaging manufacturers known for CSR in waste management) and invited them to be pilot sponsors. Their role is to financially support the recovery of a certain quantity of plastic (say, funding the operations for X tons) in exchange for receiving the Digital Recovery Certificates and Plastic Credits issued. These early corporate users are acquired through direct engagement at senior levels, facilitated by UNDP's networks and the Accelerator program. We highlight how participating in the pilot gives them a first-mover advantage: they get to shape the solution to fit their reporting needs and can publicize their involvement as an innovation in their ESG portfolio. Since Plastiks already operates internationally, we also leverage success cases (e.g., a multinational that used Plastiks to trace several million kilograms in another country, improving their ESG ratings) to instill confidence that this is a credible platform. For corporate users, the user experience is largely via the dashboard and reporting interfaces. We ensure during acquisition that we understand their data requirements (for EPR filings, for annual sustainability reports, etc.) and configure the dashboards to deliver those metrics. A key part of acquiring and keeping corporate users is the quality of data: thanks to the blockchain and our verification steps (photo, GPS, documentation for each batch), the data has high integrity, which we know is paramount for them. We offer hands-on support in analyzing and interpreting the results so that companies can seamlessly integrate the figures into their CSR reports or compliance submissions, making Plastiks not just a data provider but a solution partner. This white-glove treatment for initial corporate users is crucial for word-of-mouth in the business community; if we make champions out of them, they will advocate for Plastiks in industry forums.

4.5 Public Sector and Institutional Users

Finally, acquiring the buy-in of local governments and national bodies is addressed. While they may not be "users" in a conventional sense (not paying customers), their participation is critical to scale. We onboard city officials by offering them government-specific logins to the Plastiks dashboard where they can see city-level aggregated data – for example, how much plastic was collected in their jurisdiction, which can feed into their Swachh Bharat (Clean India) mission metrics or SDG local monitoring. The convenience of having live data is a selling point. We will train designated ULB staff on using the system for monitoring and even for issuing digital

acknowledgments or incentives to the waste picker force. For instance, a city commissioner might use the data to decide which waste picker groups to formalize or reward in municipal programs. By making these officials passive users of the data, we increase the chances of institutionalization (they might adopt Plastiks as an official tool in the future). At the national level, we engage the CPCB (Central Pollution Control Board) which runs the EPR portal. Demonstrating interoperability – that our system can feed into theirs – is an acquisition of a different sort: it's about getting regulatory endorsement. If achieved, that essentially mandates many users to come on board in order to comply with the law.

In conclusion, our user acquisition strategy is multifaceted and meticulously tailored to each stakeholder. It combines grassroots mobilization with high-level partnerships, ensures low barriers to entry and immediate value, and builds trust through community engagement and reliable performance. By the end of the pilot, we aim to have a critical mass of each user type – a vibrant base of waste pickers and aggregators, a cohort of recyclers, and a handful of committed companies and supportive officials – which together form a self-reinforcing network that will propel Plastiks into its full-scale launch.

5. Go-to-Market Execution and Pilot Implementation

Our go-to-market strategy is centered on leveraging the pilot as a springboard for broader adoption, using a structured approach of testing, feedback, iteration, and strategic partnerships. The Milestone 5 phase coincides with moving from prototype to a live pilot in the field, and we have laid out a clear execution plan for this rollout in India.

5.1 Pilot Design

The pilot is designed as a controlled launch in real operational environments. It focuses on two urban locations (e.g., one in North India – Greater Noida in NCR – and one in West India – Pune district) identified for their active recycling ecosystem. Each location serves as a micro-market where we can refine the model. The design includes setting up collection points enabled with Plastiks technology: for example, in Pune we collaborate with an existing material recovery facility where waste pickers

already bring plastics; we equip it with our QR scanning app and train the staff. In Greater Noida, we partner with the municipal waste concessionaire to introduce Plastiks at a few community collection centers. By embedding in existing infrastructure, we minimize disruption and gain immediate traction (the volumes were already flowing; we are now capturing the data). The pilot phase runs for several months, during which we aim to track at least hundreds of tons of plastic to generate meaningful data. We define clear metrics for success: number of waste pickers enrolled, kilograms of plastic recorded on-chain, number of Plastic Credit certificates issued, and feedback satisfaction from users.

5.2 Feedback Loop and Iteration

Integral to the pilot is a robust feedback loop. We deploy a field team (including members from Plastiks and UNDP Accelerator) who regularly interact with participants. Waste pickers and scrap shop owners are encouraged to share their experiences – what difficulties they face, any misunderstandings, or suggestions – through weekly debriefings or even a simple feedback form (helped by NGO partners for those who are illiterate). For instance, if waste pickers report that the QR cards get damaged easily in the field, we respond by making them more durable (laminating them or offering QR stickers as backup). If collectors find the app interface confusing in parts, we note those UI pain points and swiftly update the software. Because the system is digital, we can observe usage patterns in real time as well: if certain transactions are not being completed or data seems inconsistent, our team investigates and fixes any technical bugs or provides additional training. This agility in response ensures that by the end of the pilot the solution is truly adapted to local conditions. We treat the pilot communities as co-creators; their feedback directly shapes features like language options in the app, the format of reports, and the nature of incentives. Additionally, a formal pilot advisory committee meets monthly – including representatives from the waste picker group, the local NGO, the recycler, a city official, and a corporate partner – to review progress and advise on adjustments. This inclusive governance of the pilot ensures transparency and alignment with everyone's expectations. It also builds goodwill, as each stakeholder feels heard and invested in the solution.

5.3 Go-to-Market Partnerships

While the pilot provides proof, the go-to-market plan ensures that as we prove, we also position for growth. We have secured key partnerships as distribution and scale channels. For example, UNDP's collaboration with the Ministry of Social Justice (which oversees welfare for informal workers) is leveraged to potentially integrate Plastiks into the Ministry's programs. Post-pilot, this could mean when the Ministry issues new **"Safai Mitra" ID cards nationally, they could consider incorporating QR codes compatible with Plastiks**, immediately creating a pathway for millions of workers to join the platform. Similarly, our partnership with industry bodies (like FICCI and the Responsible Sourcing Network mentioned in UNDP's project) gives us a platform to pitch Plastiks as the recommended tool for EPR fulfillment. As part of the go-to-market, we prepare a **toolkit for partners** – e.g., a city government toolkit that describes how to adopt Plastiks in their waste management system, and a corporate toolkit for how to sponsor credits or integrate their CSR with Plastiks. These toolkits, backed by pilot data and case studies, will be disseminated through UNDP's channels and our own networks at the conclusion of the pilot.

5.4 Marketing and Outreach

On the execution front, a targeted marketing campaign accompanies the pilot. We deliberately generate local media coverage – inviting journalists to see the system in action at the pilot sites, resulting in news stories that highlight the innovative approach to waste management and the human stories of waste pickers. This media attention not only boosts public awareness (supporting branding) but also acts as indirect marketing to other cities and companies. We also plan demonstration events: for instance, an "Open House" day where government officials from neighboring municipalities or CSR heads from companies are invited to the pilot site to witness firsthand how Plastiks works. Seeing the QR scans, the digital dashboard, and talking to waste pickers who have benefitted can convert skeptics into adopters. These events are part of our go-to-market strategy to seed demand beyond the pilot locations.

5.5 Testing Structure and Risk Mitigation

The pilot serves to test not only the technology but also the assumptions in our business model. We test the pricing model – currently set at USD 0.01 per kilogram – during this controlled rollout. For the pilot participants, the fee is subsidized or waived (since we have grant funding), but we simulate how revenue would be generated by tracking the volumes and calculating what would have been earned. This allows us to validate that $0.01/kg is viable: initial data indicates that even at scale, this fee is a tiny fraction of typical waste management costs, hence easily acceptable to paying clients, while at sufficient volume it sustains the platform. We test user behaviors, adoption rates, and the effectiveness of incentives, ready to tweak our approach for the larger go-to-market. Risks like technology failure, data inaccuracies, or user drop-off are mitigated through the tight feedback loop described. Additionally, fraud prevention measures are tested: we verify that features like photo GPS evidence and blockchain hashing indeed catch any anomalies (for example, if someone tries to scan the same QR twice for the same material, the system flags it). By pilot's end, we will have a refined, proven product and playbook.

In conclusion, the go-to-market execution built around the pilot ensures we are not launching blindly, but based on evidence and refined methods. The pilot is both a proving ground and a marketing vehicle. Through careful implementation, continuous improvement via feedback, and proactive outreach to new partners, we lay the groundwork for Plastiks to enter the broader market confidently. The result will be a replicable model of traceable plastic recovery that we can roll out across other Indian cities and beyond, supported by the credibility of a successful pilot under UNDP's guidance.

6. Post-Accelerator Sustainability and Scaling Plan

As the Plastiks x UNDP India pilot concludes its accelerator phase, we are focused on ensuring the initiative not only survives but thrives as a sustainable venture with a clear path to scale. This section outlines how we will transition from a donor-supported pilot to a self-sustaining operation integrated with larger systems, and how we plan to scale impact geographically and financially.

6.1 Financial Sustainability through Market Mechanisms

A cornerstone of sustainability is our pricing model of USD 0.01 per kg of plastic recorded. This seemingly modest fee is designed to be affordable for stakeholders while generating steady revenue at scale. During the accelerator-supported pilot, operational costs were covered by grant funds, but moving forward, revenue from the platform will begin to offset expenses. For example, if a medium-sized city uses Plastiks to trace 10,000 tons of plastic annually (10 million kg), the platform would earn $100,000 from that activity – enough to cover technical maintenance and local support staff for that city. Our model anticipates multiple revenue streams on top of the per-kg fee: Plastic Credit transactions provide a commission (whenever a company purchases credits generated through the platform, Plastiks takes a small percentage). Additionally, premium services like customized compliance reports or API integration for big corporates can be offered at an added cost. We have kept the base fee low to encourage adoption (cost should not be a barrier for anyone to join), counting on the large volumes of waste in India and the value-added services for profitability. It is worth noting that the economics of plastic recovery credits are evolving – similar to carbon credits, prices can fluctuate based on demand. At $0.01/kg, we have a baseline, but if our verified Plastic Credits attract higher market prices (some voluntary markets peg plastic credits at $0.10 to $0.20 per kg depending on the social impact), part of that upside can translate to higher revenues (shared with waste pickers and project sustainment). The key is that by linking our revenue to the measurable impact (kg recycled), we ensure a virtuous cycle: the more impact we create, the more financial resources we have to continue creating impact.

6.2 Institutional Partnerships and Integration

Post-accelerator, Plastiks will deepen its integration with public infrastructure and UNDP priorities. We are aligning our platform as a digital public good that can be scaled with government support. Discussions are underway to integrate Plastiks with India's national EPR portal and potentially the Ministry of Environment's waste management dashboards. If Plastiks becomes an officially recommended tool for EPR compliance (for instance, recognized by the Central Pollution Control Board), it gains long-term stability – essentially becoming part of regulatory infrastructure. In

parallel, we look to integrate with municipal systems: our open APIs can allow city waste management apps or Swachh Bharat mission databases to pull Plastiks data, embedding our solution in day-to-day public services. UNDP's role post-accelerator often shifts to advisory and advocacy; we will continue working with UNDP India to advocate for policy frameworks that support traceable recycling. This includes contributing to guidelines and standards – for example, working with the Bureau of Indian Standards or other bodies on standardizing QR code usage for waste traceability, which would enshrine compatibility with our approach at a national level. By being part of the committees or working groups (with UNDP's facilitation) that design these standards, we ensure that the resulting policies favor solutions like ours and potentially even create mandates that require our kind of system (similar to how electronic invoicing became mandatory in certain sectors, which locked in demand for those solutions).

6.3 Scaling Model - Hub-and-Spoke and Knowledge Transfer

Scaling will proceed in a hub-and-spoke model initially across India, and then internationally. Within India, we will use the pilot cities as centers of excellence. The local teams and champions developed in Pune and Noida will help train and set up the platform in new cities. Our scaling plan identifies priority geographies: major metro areas with significant plastic waste and active informal sectors (e.g., Delhi beyond Noida, Mumbai, Bangalore, Chennai) and also secondary cities where the need for formalization is high (e.g., tourist hotspots or coastal cities dealing with plastic leakage). We plan to partner with state governments and regional UNDP offices to replicate the model. Each new location will adapt the blueprint from the pilot, with adjustments for local context (language, specific waste streams, etc.). We'll also develop a "Plastiks Implementation Guide" as a product of the pilot, which can be shared with any interested city or organization – essentially a manual on how to onboard informal workers, set up the tech, and run the system, distilled from our experience. By making this knowledge accessible (possibly as part of UNDP's knowledge sharing under SDG Accelerator), we invite wider adoption even beyond our direct reach.

Internationally, Plastiks is already engaged in other countries (as evidenced by the Armenia pilot and presence in 20+ countries). The India pilot's success will be a

powerful case study to showcase globally, especially for countries with large informal waste sectors. UNDP's Alternative Finance Lab and the Blockchain Accelerator program can facilitate introductions to other UNDP country offices. We foresee a scenario where Plastiks becomes a recurring solution in multiple countries' SDG portfolios – a scalability that is not just technical but also institutional, backed by UNDP's global network. This cross-pollination also adds to sustainability: the more diverse our operations, the more resilient the organization (learning from different contexts, spreading financial risk across markets, etc.).

6.4 Sustainability of Impact and Community

Beyond financial and operational sustainability, we measure success in the sustained social and environmental impact. One risk with pilot projects is that benefits to communities fade if the project ends abruptly. To guard against this, the scaling plan includes institutionalizing community gains. For example, any waste picker who obtained a digital ID and was enrolled in social protection during the pilot will continue to have that status – we coordinate with local authorities to ensure no one is "deregistered" when the pilot concludes, because by then the local authority ideally sees the value in maintaining those IDs and services (indeed, our data can help them justify continuing support by showing improved waste management metrics). The pilot's waste picker participants could also form the nucleus of cooperatives or associations that persist. We have encouraged the formation of local committees among the waste pickers for self-organization. These groups, empowered with data about their own contributions (e.g., "together we collected 500 tons in this ward last year"), are in a stronger position to negotiate for their rights and pricing with buyers even independent of Plastiks. In essence, the platform has given them information leverage that remains a lasting asset.

6.5 Alignment with SDGs and Long-Term Vision

Finally, sustainability is reinforced by aligning with long-term global goals that are not going away. By contributing directly to SDG 12 (Responsible Consumption and Production) – measured through increased recycling rates (indicator 12.5.1) – and SDG 8 (Decent Work and Economic Growth) – through the formalization of informal work (indicator 8.3.1) – and tangentially to SDG 13 (Climate Action) – by reducing

open burning and plastic pollution, integrating into climate policy (indicator 13.2.1) – we ensure that our project remains relevant to governments and international agencies in the years ahead. Countries and corporations are committing to circular economy and ESG targets in the 2030 and 2050 timelines; Plastiks provides a concrete mechanism to achieve and measure progress on those. This alignment means we can tap into continued streams of climate finance and SDG-aligned funding. For instance, a government may fund Plastiks rollout as part of its SDG action plan, or a climate fund might invest in scaling plastic waste avoidance as a mitigation strategy (plastic production and disposal are significant sources of greenhouse gas emissions, so improving waste management has climate benefits).

Our post-accelerator plan can be summarized as consolidate, institutionalize, and expand. We consolidate the gains and insights of the pilot into a stable model; institutionalize the solution by embedding it in policies, systems, and communities; and expand to reach a larger scale of impact. By doing so, Plastiks x UNDP India will transform from a time-bound pilot into a lasting program that continues to thrive, scale, and innovate, helping India (and other countries) achieve a more sustainable and inclusive management of plastic waste well beyond the life of the accelerator.